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Model: CatBoost Classifier

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CardioRisk AI

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Cardiovascular Disease Prediction using CatBoost

Project Overview

CardioRisk AI is a machine-learning system trained to predict cardiovascular disease (CVD) from clinical and lifestyle features. Cardiovascular disease is the leading cause of death worldwide — early, data-driven detection can save lives.

Pipeline summary:

Dataset: 70,000 row cardiovascular health records (Kaggle)

Cleaning: removed duplicates, outliers in height/weight/blood pressure

Feature engineering: age_years: from days, BMI calculation

Models trained: 7 classifiers compared; CatBoost selected as best

Best ROC AUC: ~ 0.80

Features used: age_years, gender, height, weight, BME, ap_hi (systolic BP), ap_lo (diastolic BP), cholesterol, gluc, smoke, alco, active

Total Patients

68,608

With CVD

33,942

↑ 49.5%

Healthy

34,666

↓ 50.5%

Dataset Quick Peek

id	age	gender	height	weight	ap_hi	ap_lo	cholesterol	gluc	smoke	alco	active	cardio	age_years	BMI	
0	0	18393	2	368	62	110	80	1	1	0	0	1	0	50	21.9671
1	1	20228	1	156	85	140	90	3	1	0	0	1	1	55	34.9277
2	2	18807	1	165	64	130	70	3	1	0	0	0	1	51	23.5078
3	3	17623	2	369	82	150	100	1	1	0	0	1	1	48	28.7185
4	4	17474	1	156	56	100	60	1	1	0	0	0	0	47	23.0112
5	8	21914	1	151	67	120	80	2	2	0	0	0	0	60	29.3947
6	9	22113	1	157	93	130	85	3	1	0	0	1	0	60	37.7287
7	12	22584	2	178	95	130	90	3	3	0	0	1	1	61	29.9836
8	13	17668	1	158	71	110	70	1	1	0	0	1	0	48	26.441
9	14	19814	1	164	68	110	60	1	1	0	0	0	0	54	25.2825

CardioRisk AI

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Dataset Statistics

	count	mean	std	min	25%	50%	75%	max
id	68608	49976.57	28846.37	0	23005.75	50018	74871.25	99999
age	68608	19464.92	2467.91	10798	17658	19702	21325	23713
gender	68608	1.35	0.48	1	1	1	2	2
height	68608	164.41	7.91	120	159	165	170	207
weight	68608	74.12	14.3	30	65	72	82	200
ap_hi	68608	126.68	16.68	80	120	120	140	140
ap_lo	68608	81.31	9.43	40	80	80	90	100
cholesterol	68608	1.36	0.68	1	1	1	2	3
gluc	68608	1.23	0.57	1	1	1	1	3
smoke	68608	0.09	0.28	0	0	0	0	1

Missing Values

Feature	Missing Count
5 ap_hi	0
6 ap_lo	0
7 cholesterol	0
8 gluc	0
9 smoke	0
10 alco	0
11 active	0
12 cardio	0
13 age_years	0
14 BMI	0

Feature Distribution

Select feature

weight

weight

Count

weight

weight

cardio

Select feature

weight

weight

Count

weight

weight

cardio

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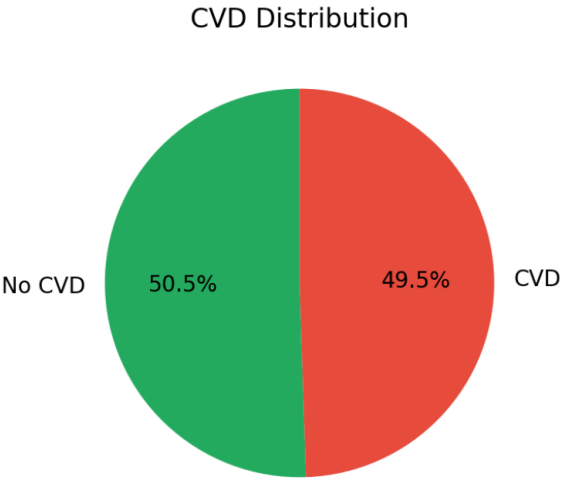
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Target Distribution



No CVD
50.5%

CVD
49.5%

The dataset is well-balanced, ideal for classification tasks.

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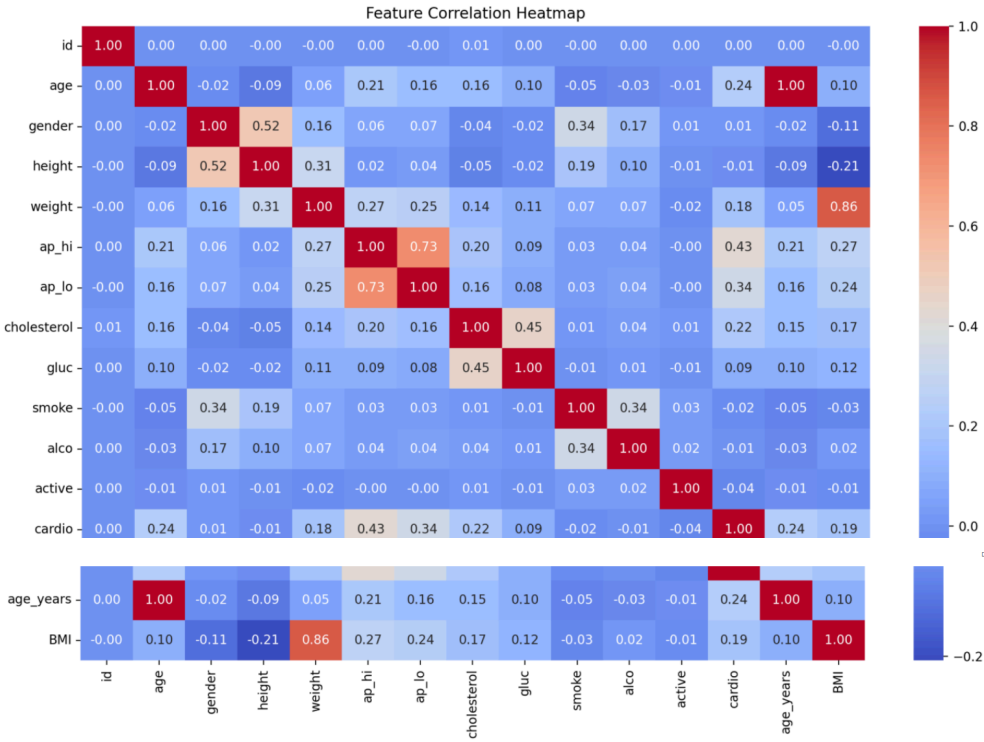
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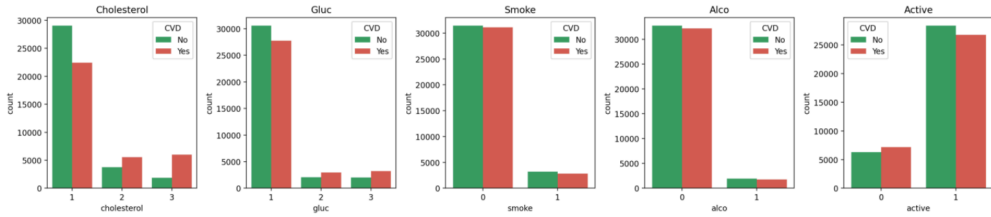
Model: CatBoost Classifier

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Correlation Heatmap



Lifestyle Factors vs CVD



- Key EDA Findings:
- Older patients show significantly higher CVD prevalence.
| 2 | Elevated systolic blood pressure is strongly associated with CVD. |
| 3 | Higher BMI correlates with increased cardiovascular risk. |
| 4 | Cholesterol level is a powerful predictor. |
| 5 | Physical activity appears protective. |

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CardioRisk AI

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Model Performance Comparison						
Model	Accuracy	Precision	Recall	F1 Score	ROC AUC	
0 CatBoost	73.558000	75.640000	68.660000	71.980000	0.804200	
1 Gradient Boosting	73.558000	75.670000	68.710000	72.620000	0.802900	
2 Decision Tree	73.300000	76.200000	66.930000	72.270000	0.795900	
3 AdaBoost	72.880000	76.490000	65.220000	70.410000	0.795400	
4 Logistic Regression	72.728000	75.580000	66.250000	70.610000	0.792800	
5 Random Forest	71.378000	71.890000	69.840000	70.650000	0.770700	
6 KNN	69.528000	69.660000	68.940000	69.840000	0.744900	

ROC AUC Comparison

Model Comparison — ROC AUC (CatBoost = Best)

Model	ROC AUC Score
KNN	0.7449
Random Forest	0.7707
Logistic Regression	0.7928
AdaBoost	0.7954
AdaBoost	0.7954
Decision Tree	0.7959
Gradient Boosting	0.8029
CatBoost	0.8042

Confusion Matrix — CatBoost

Confusion Matrix

Actual \ Predicted	No CVD	CVD
No CVD	5432	1501
CVD	2128	4661

ROC Curve

CatBoost AUC=0.8042

Feature Importance — CatBoost

Feature Importances (CatBoost)

Feature	Importance
ap_hi	~0.95
age_years	~0.85
cholesterol	~0.75
BMI	~0.55
weight	~0.55
ap_lo	~0.55
gluc	~0.45

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Feature Importances (CatBoost)

Feature	Importance
ap_hi	37
age_years	21
cholesterol	13
BMI	6
weight	5.5
ap_lo	5.5
gluc	4
height	3.5
active	2.5
gender	1.5
smoke	1.5
alco	1.5

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Patient Risk Assessment

Enter patient details below to predict cardiovascular disease risk.

Biometrics

Age (years)

18

Gender

Female

Height (cm)

178

Weight (kg)

75

Calculated BMI

25.95

Overweight

Blood Pressure & Lab

Systolic BP (ap_hi)

178

Diastolic BP (ap_lo)

98

Cholesterol

Normal

Glucose

Normal

Lifestyle

Smoking?

No

Alcohol?

No

Physically active?

No

Predict Cardiovascular Risk

Prediction Result

LOW RISK of Cardiovascular Disease - Probability: 15.4%

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CVD Risk Score: 15.4%

Risk	Predicted Probability
Low Risk	0.154

Disclaimer: This tool is for educational purposes only and does not replace professional medical advice.